

VisionGPT Multimodal System Technical Overview

1. System Architecture

VisionGPT is an enterprise multimodal retrieval-augmented generation framework. It integrates local language m

2. Vision Processing Module

The vision module uses Microsoft Florence-2-base model to generate detailed image captions and video frame c

3. Vector Indexing

Embeddings are computed using BAAI/bge-small-en-v1.5 producing 384-dimensional dense vectors stored in FA

4. Speech Recognition Engine

Audio transcription is performed by Faster-Whisper base model operating on CPU with int8 quantization. It generates a transcript of the audio input.

5. Language Model Synthesis

The RAG synthesis phase is executed by local Ollama serving qwen2.5:3b with standalone query rewriting.